

5.2 Energy

Born–Haber cycles are used as a theoretical model to illustrate the energy changes associated with ionic bonding.

Entropy and free energy are then introduced as concepts used to predict quantitatively the feasibility of chemical change.

Redox chemistry permeates chemistry and the introductory work in Module 2 is developed further within this section, including use of volumetric analysis for redox titrations and an introduction of electrochemistry in the context of electrode potentials.

5.2.1 Lattice enthalpy

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
Lattice enthalpy	
(a) explanation of the term <i>lattice enthalpy</i> (formation of 1 mol of ionic lattice from gaseous ions, $\Delta_{\text{LE}}H$) and use as a measure of the strength of ionic bonding in a giant ionic lattice (see also 2.2.2 b–c)	Definition required.
Born–Haber and related enthalpy cycles	
(b) use of the lattice enthalpy of a simple ionic solid (e.g. NaCl, MgCl ₂) and relevant energy terms for: (i) the construction of Born–Haber cycles (ii) related calculations	M2.2, M2.3, M2.4, M3.1 Relevant energy terms: <i>enthalpy change of formation, ionisation energy, enthalpy change of atomisation and electron affinity.</i> Definition required for first ionisation energy (see also 3.1.1 c) and enthalpy change of formation (see also 3.2.1 d) only. HSW2 Application of conservation of energy to determine enthalpy changes.
(c) explanation and use of the terms: (i) <i>enthalpy change of solution</i> (dissolving of 1 mol of solute, $\Delta_{\text{sol}}H$) (ii) <i>enthalpy change of hydration</i> (dissolving of 1 mol of gaseous ions in water, $\Delta_{\text{hyd}}H$)	Definitions required. Details of infinite dilution not required.
(d) use of the enthalpy change of solution of a simple ionic solid (e.g. NaCl, MgCl ₂) and relevant energy terms (<i>enthalpy change of hydration and lattice enthalpy</i>) for: (i) the construction of enthalpy cycles (ii) related calculations	M2.2, M2.3, M2.4, M3.1 HSW2 Application of conservation of energy to determine enthalpy changes.

- (e) qualitative explanation of the effect of ionic charge and ionic radius on the exothermic value of a lattice enthalpy and enthalpy change of hydration.